DSA REFERENCE

Hash Maps and Counting

Recognition cues, invariants, code templates, and 5-8 referenced problems.

Store a compact summary of values already seen so a repeated lookup replaces a nested scan.

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STATUS HUMAN REVIEW REQUIRED

DSA / HASH-MAPS

DSA

Hash Maps and Counting

CORE IDEA

Store a compact summary of values already seen so a repeated lookup replaces a nested scan.

Recognition cues

+ pair or complement lookup
+ frequency or grouping
+ first or last occurrence
+ membership with order-independent access

Implementation sequence

+ choose the minimum state needed
+ query before updating when self-matches are invalid
+ define duplicate behavior
+ state expected O(1) lookup assumptions

Invariant

The map describes exactly the processed prefix, or exactly the active window when entries are removed.

Worked reasoning

For Two Sum, store value to index; for each value x, check target - x before inserting x.

Complexity

Usually O(n) time and O(n) space; adversarial hashing caveats are rarely the interview focus.

Failure modes

- [ ] updating before querying
- [ ] losing duplicate indices
- [ ] using a set when counts matter
- [ ] forgetting window removals

Practice ladder

Hash Maps and Counting: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Two Sum**	State the cue, invariant, edge case, and proof before implementation.
2	**Group Anagrams**	State the cue, invariant, edge case, and proof before implementation.
3	**Top K Frequent Elements**	State the cue, invariant, edge case, and proof before implementation.
4	**Longest Consecutive Sequence**	State the cue, invariant, edge case, and proof before implementation.
5	**Valid Sudoku**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

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